Plant-CellFM: an auditable foundation framework for cross-species plant single-cell annotation and context adaptation

Abstract

Plant single-cell and single-nucleus atlases are expanding across model plants, crops and non-model species, yet annotation transfer remains confounded by gene-identifier differences, open cell-state vocabularies and mixed evaluation protocols. We present Plant-CellFM, a plant expression modelling framework that treats annotation as an auditable evidence chain rather than a single classifier score. The frozen corpus profile contains 272,732 cells, 209,405 genes, five profiled species, nine public datasets, 31 samples and 34 raw labels; the separate strict evaluation panel contains 3,964 aligned cells from eight held-out species. Plant-CellFM combines a 256-dimensional four-layer encoder, an explicit gene-identifier and orthology contract, LoRA adaptation interfaces, hierarchical annotation outputs, ranked marker candidates and a runtime annotation head. In the primary nested leave-species protocol, target labels are excluded from fitting, rule selection and calibration. The result is 39.96% all-cell accuracy, 55.90% source-label coverage, 71.48% accuracy and 0.2817 macro-F1 on the covered-label subset. A separate global context-sensitivity analysis, reported outside the primary protocol, reaches 42.36% all-cell accuracy with the same frozen panel and denominator; its rule-selection boundary remains explicit. Labelled target-species adaptation gives a reproducible dose response from 59.21% to 75.89% mean query all-cell accuracy as support increases from 8 to 64 cells per species across ten disjoint support-query draws. We further provide a label-free external root execution audit, a marker-candidate resource and three author-labelled adaptation cases. In GSE270140/GSM8335426, a LoRA-mode context adapter achieves 83.97% fine accuracy and 84.47% macro-F1 on 2,352 locked test cells, while a pre-registered three-state vascular mapping increases semantic accuracy from 2.02% for the frozen base checkpoint to 90.93% after adaptation. In a barcode-non-overlap allopolyploid wheat-root case (GSE270342), author orthogroups retain 76.33% of source UMI counts and a fixed-split LoRA adapter reaches 62.25% accuracy and 0.6660 macro-F1 across 1,433 locked author-labelled cells. Finally, the source-pinned GSE297576 Sorghum bicolor root atlas is absent from the frozen five-species corpus: frozen inference reaches 14.56% accuracy and 0.1083 macro-F1 on 14,909 predeclared comparable cells, whereas a rank-8 LoRA adapter trained on two libraries, selected on a third and tested once on a sealed fourth library reaches 76.02% 27-state accuracy and 0.7535 macro-F1 across 4,150 cells. On the same 3,549 broad-identity test cells, accuracy rises from 14.79% to 84.98%. The high-scoring Sorghum result is explicitly target-species adaptation, not zero-shot or independent model ranking. Plant-CellFM therefore provides a reproducible route for separating strict transfer, external open-set screening, target-context adaptation and deployment annotation in plant atlas analysis.

On the same locked wheat-root test, an official scPlantLLM checkpoint loaded without state-key discrepancies reached 0.2001 macro-F1 through a frozen encoder plus train-only centroid readout, 0.2998 macro-F1 after final-block-plus-new-head partial adaptation and 0.4588 macro-F1 after full-backbone-plus-new-head adaptation. Both adapted prediction records replay exactly from their recorded checkpoints. This reference shares the prepared object, author mapping and test barcodes but is presented only as a same-study adaptation comparison, not independent validation, strict transfer or a compute-budget-matched rank.

Introduction

Plant atlases now span organs, genotypes, developmental stages and phylogenetic lineages at cell resolution. Their value depends on whether cell identities can be transferred and compared without silently changing the meaning of a label. In practice, species differ in gene identifiers and orthology, organs differ in state composition, and target datasets often contain labels that are absent from a source vocabulary. A model can also be scored under fundamentally different information boundaries: strict transfer excludes target labels, few-shot adaptation uses a labelled support set, and a deployed head may already contain a fixed output vocabulary. These are distinct questions and should not be compressed into a single accuracy statement.

Plant-CellFM addresses this problem with a protocol-aware design. The framework records the provenance of input matrices, aligns genes through exact identifiers or an explicit orthology map, resolves species adapters, emits hierarchical annotations and marker candidates, and attaches the evaluation protocol to the reported result. The scope is plant-general: Saussurea involucrata is a future target-species use case rather than a model boundary. The current package does not claim a completed Snow Lotus single-cell atlas because no reusable Snow Lotus single-cell matrix is included.

Here we establish a submission-scale evidence package with a frozen corpus profile, an 8-species nested strict-transfer benchmark, a label-integrity companion analysis, repeat-sampled target-species adaptation, matched internal checkpoint comparisons, a matched official third-party reference, a label-free external root execution case, a literature-anchored root candidate resource, a secondary-root context adapter, a provenance-controlled allopolyploid wheat adaptation case and a source-pinned Sorghum external-screen and sealed-library adaptation case. Each quantitative result is paired with source data, a declared denominator and a record of what information entered fitting or selection.

Results

A frozen corpus profile and explicit input contract separate scope from performance

The frozen Plant-CellFM profile comprises 272,732 cells and 209,405 genes from five profiled species, nine public datasets and 31 samples (Fig. 1). The profile is deliberately distinct from the strict evaluation panel, which contains 3,964 aligned cells drawn from eight held-out species. This separation prevents historical catalogues, evaluation-only species or registered adapter names from being misrepresented as cells in the frozen corpus.

The encoder uses four layers and a 256-dimensional cell representation. Inputs first pass through an exact gene-identifier contract; when genome versions or species require it, an orthology map is supplied as a visible artifact. The model exposes cell embeddings, hierarchical labels, ranked marker candidates and adapter-resolution metadata. This interface is a reusable method component, not a claim that every named plant has already received empirical validation.

Nested leave-species transfer retains the open-set denominator

The primary cross-species experiment is the v17 nested metadata gate. For every outer held-out species, the decoding rule is selected only with inner leave-species evaluation on source species. Target labels do not enter encoder fitting, rule selection, threshold calibration or post-hoc error correction (Fig. 2 and Extended Data Fig. 2). The resulting all-cell accuracy is 39.96% over all 3,964 test cells. Source-label coverage is 55.90%; the accuracy and macro-F1 restricted to the covered-label subset are 71.48% and 0.2817, respectively.

These quantities describe complementary, rather than competing, aspects of transfer. All-cell accuracy counts cells carrying target labels absent from the source label vocabulary as errors. The conditional measures describe recognition among labels that the source could in principle express. Both are shown because presenting only conditional accuracy would remove the central open-set challenge from the denominator.

The v18 label-integrity companion analysis evaluates a second, pre-specified question. Labels beginning unknown, unknow or unannotated are retained as audit records but do not enter identity fitting or identity scoring. This yields 2,324 explicit-identity cells and 1,640 audit-only labels. The companion analysis is not a replacement for v17; it makes the influence of non-identity public labels inspectable rather than allowing them to create a misleadingly clean or misleadingly poor score.

We separately examined whether context could reduce cross-species error on the frozen panel using a context-aware species-transfer calibration (phylo_organ_gate_v1). The gate uses organ and phylogenetic support estimated from source species: it retains expression-based transfer when informative same-family support is available, and otherwise falls back to an organ prior. On the same 3,964 cells, exact-label denominator and 55.90% source-label coverage, all-cell accuracy rises from 23.64% for the centroid baseline through 30.10% for expression STC and 31.84% for neural STC to 42.36% for the context-aware gate (Fig. 2e). Covered-label accuracy reaches 75.77% and covered-label macro-F1 is 0.3045. Because configuration selection aggregates outer-fold information, this is a global context-sensitivity analysis; it is not a replacement for the nested v17 primary result. It is retained to identify an implementable context module and to make its selection boundary inspectable.

Target-species support produces a repeatable adaptation response

Strict zero-shot transfer asks what can be predicted without target labels. Atlas construction frequently offers a different route: a small set of target cells can be labelled before the remaining cells are annotated. We therefore sampled 8, 16, 32 or 64 labelled support cells per target species, excluded them from query scoring, and repeated each support budget across ten independent draws (Fig. 3).

Mean query all-cell accuracy increased monotonically from 59.21% at 8 support cells to 67.34%, 72.30% and 75.89% at 16, 32 and 64 support cells. Query macro-F1 increased from 0.2195 to 0.4619. The panel retains the raw draws and species-by-budget values, making both variability and species heterogeneity visible. These are labelled adaptation results and are never substituted for the strict v17 zero-shot score.

Matched evaluations distinguish model improvement from comparator scope

Frozen v3 and Plant-CellFM v9 checkpoints are compared only on shared datasets, splits, gene contracts and label coverage. All-cell accuracy rises from 20.21% to 44.90% in leave-dataset evaluation and from 41.55% to 62.00% in leave-sample evaluation. In the recorded label-normalized leave-species comparison, it rises from 19.12% to 23.54% (Extended Data Fig. 3). These are matched internal checkpoint results, not a ranking against unrelated external tools.

Seurat label transfer and a cosine-centroid baseline have separate recorded executions. We additionally evaluated the official scPlantLLM checkpoint on the GSE270342 prepared object through the exact author first-target mapping and the exact 1,433-cell locked Plant-CellFM test barcode set (Extended Data Fig. 8; Tables S22-S24). The checkpoint loaded on CUDA with zero missing or unexpected state keys. A frozen-encoder, train-only centroid readout obtained 0.2107 accuracy and 0.2001 macro-F1. Freezing the first five transformer blocks and training the final block plus a new 13-class head, with epoch 4 selected solely by validation macro-F1, reached 0.3426 accuracy and 0.2998 macro-F1. Training every scPlantLLM backbone parameter plus a new 13-class head, with epoch 8 selected by the same validation-only rule, reached 0.4501 accuracy and 0.4588 macro-F1. Separate replay audits regenerated each adapted test prediction table and its metrics from the recorded checkpoint checksum. The shared object, mapping and lock make these informative executable same-study references, but their differing trainable scope and compute do not establish independent validation, strict transfer, a compute-matched comparison or a universal model ranking. scPlantAnnotate still requires authenticated execution or an official prediction export.

External root execution and literature-fixed markers provide a biological audit layer

We applied the frozen root checkpoint to the label-free Arabidopsis root matrix GSE152766/GSM4626007. The input contains 6,566 cells and 25,171 TAIR10 gene identifiers, is not listed in the frozen v4 profile, and contains no expert cell-identity field. The execution is consequently a blind inference audit, not an external accuracy experiment. The model returns 13 predicted states, confidence values and 256-dimensional embeddings (Fig. 4 and Extended Data Fig. 5).

Before analysing the matrix, we fixed six canonical root marker-identity expectations from the primary literature. Five markers, COBL9, GL2, CASP1, APL and MYB46, have the highest mean expression and detection rate in their expected predicted group; WER remains a positive but non-top signal. The phloem prediction group has four cells and that denominator is retained in the source data. Separately, the root candidate resource covers ten identities with top-20 candidates per identity, for 200 rows. Three of six literature-fixed anchors are recovered in their matching top-20 programs. These analyses support biological coherence and experimental prioritization, not external accuracy or wet-lab validation.

A secondary-root context adapter resolves author-defined states on locked cells

GSE270140/GSM8335426 is a public Arabidopsis secondary-root study that combines single-cell sequencing with lineage tracing. We used its author object only for a labelled, within-sample adaptation study. The preparation chain preserves raw feature names, cell identifiers and author annotations while canonicalizing TAIR identifiers. The resulting matrix contains 11,760 cells and 14 author-defined states.

The frozen SRP169576 root checkpoint initialized a LoRA-mode context adapter. Unique cell identifiers were assigned to a fixed seed 80/10/20 split comprising 8,232 training cells, 1,176 validation cells and 2,352 locked test cells. The best checkpoint was selected at epoch 7 using validation fine macro-F1 only. The primary held-out evaluator reports 83.97% fine accuracy and 84.47% macro-F1; a separate full-precision recheck reports 84.18% and 84.64% (Extended Data Fig. 6).

We also froze, before inference, a semantic map from compatible author labels to Phloem, Xylem and Root stele. On the same 1,885 compatible held-out cells, semantic accuracy increases from 2.02% for the frozen base checkpoint to 90.93% after adaptation, with macro-F1 0.9159. The figure retains the split, validation trajectory, all 14 classes, rare-state F1 values, source data and 3,000 fixed-seed bootstrap intervals. Because all adaptation labels originate from the same author-labelled sample, this result is a context-specific supervised adaptation. It is not used as evidence of zero-shot transfer, leave-species generalization, independent external validation or superiority over a third-party model.

A provenance-controlled wheat adapter resolves a difficult allopolyploid root atlas

We next challenged the adaptation interface with the author-labelled wheat-root atlas GSE270342, a crop case with an allopolyploid genome and non-identical gene identifiers. The author object contains 7,388 cells. An earlier exploratory strict-transfer record contained 256 rows from its CS1 replicate; exact barcode matching identified 224 cells shared with the author object. We excluded all 224 shared barcodes before frozen inference, adapter fitting or test scoring, retaining 7,164 cells. This gate removes direct reuse of the recorded exploratory cells, while the case remains explicitly a same-study re-audit rather than an independent external benchmark (Supplementary Table S20).

Direct Ensembl-style identifiers did not provide an adequate common input space. We therefore used the author-published custom PLAZA orthogroup table as an explicit many-to-many transfer contract. Of 78,115 source features, 41,987 (53.75%) resolve to the frozen checkpoint vocabulary; these features retain 76.33% of source UMI counts. Frozen prediction uses a deterministic first-target projection, and a count-conserving mean-projection sensitivity is reported alongside it rather than selected post hoc. On the locked cells carrying one of eight predeclared direct root identities, the frozen first-target diagnostic reaches 25.93% accuracy and 0.1546 macro-F1.

We then initialized a LoRA-mode wheat adapter from the frozen root checkpoint and applied a fixed seed cell-level split of 5,014 training cells, 717 validation cells and 1,433 locked test cells. Epoch 8 was selected using validation macro-F1 only. The full 13-class author-label test yields 62.25% accuracy and 0.6660 macro-F1. On the same 964 direct-root test cells used for the frozen diagnostic, accuracy rises to 56.22% and macro-F1 to 0.6847, a 30.29 percentage-point increase. The released adapter is checksum-linked to the audited training checkpoint. Supplementary Table S20 binds the barcode-exclusion gate, mapping coverage, fixed split, locked-test outcomes and release SHA256 into one reviewable record. This result demonstrates a reproducible route for species-context adaptation under a declared orthology contract; it is not a zero-shot result, an independent validation or a third-party ranking.

A source-pinned external Sorghum atlas separates frozen open-set failure from sealed-library recovery

We next used the author-labelled Sorghum bicolor root atlas from GSE297576 as a source-pinned external species case. The deterministic Seurat conversion preserves 19,316 cell barcodes, 25,464 raw UMI features and 27 author celltype annotations. Sorghum bicolor is absent from the five species in the frozen corpus profile. Its author-maintained 10-species orthogroup resource maps 15,940 of 25,464 source features to Arabidopsis targets; after preprocessing, 10,325 of 10,604 retained targets are represented by the frozen checkpoint vocabulary. The full author object, count layer, orthogroup table, conversion checksum and ontology contract are recorded before scoring (Supplementary Table S25).

Frozen inference uses no author identity labels. At the predeclared broad identity resolution, 14,909 of 19,316 cells are comparable to a frozen root output. The frozen root head reaches 14.56% accuracy and 0.1083 macro-F1 over that complete comparable denominator, and returns Unknow for 63.39% of comparable cells. This negative external screen is retained as an open-set stress test rather than omitted or relabelled. We then train a rank-8 LoRA adapter using the 13,620 cells of libraries OUGHX and OWGSC, choose epoch 10 using 1,546 cells of library OWGSB, and leave all 4,150 cells of library OUGHW sealed for one-time test scoring. The adapter reaches 76.02% accuracy and 0.7535 macro-F1 across all 27 author states on the sealed library. Applying the same broad ontology to both frozen and adapted predictions on the same 3,549 comparable OUGHW cells increases accuracy from 14.79% to 84.98% and macro-F1 from 0.1218 to 0.8362 (Fig. 5; Supplementary Tables

S26-S27). This result validates a practical recovery path after the external screen; it does not alter the strict leave-species headline or claim independent third-party superiority.

Methods

Data contracts, corpus profile and label handling

The corpus profile is generated directly from the frozen H5AD structure and records cells, genes, species, datasets, samples, tissues and raw labels. Strict evaluation records are frozen separately. Exact gene identifiers are aligned to the checkpoint vocabulary; where direct identifiers are insufficient, orthology mappings are supplied as explicit input artifacts. The v18 identity companion rule is fixed before scoring and treats unknown, unknow and unannotated labels as audit-only records.

Strict transfer and adaptation protocols

The v17 protocol uses nested source-species selection. Each outer held-out species is evaluated once, after candidate rules are ranked only using source-species inner folds. All-cell accuracy uses every held-out cell as the denominator. Source-label coverage, known-label accuracy and known-label macro-F1 are calculated and reported separately. The context-aware species-transfer calibration (phylo_organ_gate_v1) is a separate global sensitivity analysis over the frozen panel: it records the same denominator and uses organ and phylogenetic support, but its configuration is selected with aggregated outer-fold information and is therefore not promoted to a nested primary result. Few-shot adaptation samples labelled support cells by species, keeps support and query cells non-overlapping, and aggregates ten independent draws per support budget.

Supervised secondary-root, wheat-root and Sorghum-root adapters

The GSE270140 adapter uses configs/gse270140_secondary_root_lora_adapter_4070.yaml: a 256-dimensional, four-layer encoder initialized from SnowLotus_CellFM_SRP169576_annotation_1024_best.pt, LoRA rank 8, class-balanced supervised loss, a maximum of ten epochs and a fixed group-random split by unique cell ID. The run was executed with CUDA on an NVIDIA GeForce RTX 4070 Laptop GPU. The released adapter checkpoint is byte-checked against the checkpoint used for the audit. Configuration, source acquisition, archive inventory, label map, test predictions, per-class scores and semantic recovery tables are versioned in the release tree.

The GSE270342 wheat adapter uses configs/gse270342_wheat_root_lora_adapter_4070.yaml. The author Seurat object is exported as a cells-by-genes count matrix while preserving cell identifiers and author annotations. Exact CS1 barcode overlap with the historical exploratory strict-transfer record is removed before data preparation. The author orthogroups.csv relation table is retained as the transfer artifact; it is many-to-many, and the deterministic first aggregation used for frozen diagnosis is exposed separately from the count-conserving mean sensitivity. LoRA training uses the same 256-dimensional four-layer root checkpoint, rank 8, class-balanced supervised loss, ten epochs and a fixed group-random 5,014 / 717 / 1,433 cell split. The promotion script copies the selected training checkpoint into the versioned models/ registry only after byte-level equality is established. The audit then checks the release checksum, fixed test cell identifiers, overlap exclusion, full 13-class test and matched eight-state direct-root comparison.

For GSE297576, the author RDS RNA count layer is converted deterministically to a cells-by-genes H5AD while retaining exact cellBC order, celltype annotations and author UMAP coordinates. The author JGI 10-species orthogroup table is used as a many-to-many Sorghum-to-Arabidopsis transfer contract; inference records the deterministic first-target projection and its feature coverage. The frozen root head is executed before author labels are joined for scoring. A separate rank-8 LoRA adapter uses explicit leave-library partitions: OUGHX and OWGSC for training, OWGSB for validation-only epoch selection and OUGHW as a sealed test library. The released adapter is byte-matched to the audited training checkpoint. The audit confirms atlas and prediction checksums, source-species absence from the frozen profile, exact cell identity, disjoint libraries, validation-only epoch selection, all 27 raw author states and the matched broad-identity recovery calculation.

Statistics, figures and release audit

The strict-transfer, secondary-root, matched wheat-root and sealed-library Sorghum figures use fixed-seed nonparametric bootstrap resampling only to visualize uncertainty conditional on their frozen test populations.

Every quantitative visual panel has a TSV source-data file. All v6 and v7 figures are exported as editable SVG and PDF, PNG preview and 600-dpi TIFF. The v7 audit checks file presence, editable SVG text, source-table presence, 600-dpi raster output, visible evidence-tier boundary language and panel labels for the Sorghum page. The evidence audits also check frozen strict metrics, released adapter checksums, wheat overlap exclusion and fixed split, and the scPlantLLM official-checkpoint, partial- and full-adapter checksums and prediction-replay contracts. They report technical readiness but do not substitute a subjective readiness score for editorial or peer review.

Data and Code Availability

The repository, source code, v7 manuscript, model card, evidence ledger, figures, source data and reproducibility records are available at https://github.com/ahvsjags/SnowLotus-CellFM on branch agent/remote-pipeline-20260728. The GSE270140, GSE270342 and GSE297576 adapters are tracked through Git LFS and have SHA256 1a306c4a5e21630a75a5a63d2867e86712b2da78eea3805eb5dc00b957134fd7, 597b7e425b0355ddfa81e4f5c9c63e85987d6b72fe49a841736b200ea6c2c22e and f6f3da3dbfb9eda48973c041fc07fa019b6f03229ae57f072afc054d8a52755f, respectively. Reproduce the context-adaptation and matched-reference evidence with:

```
python scripts/download_gse270140_external_validation.py python scripts/extract_gse270140_external_assets.py
python scripts/prepare_gse270140_external_validation.py python -m snowcell.cli train --config
configs/gse270140_secondary_root_lora_adapter_4070.yaml --device cuda python
scripts/audit_gse270140_secondary_root_adapter.py python scripts/render_v5_secondary_root_adapter_figure.py
python scripts/download_gse270342_author_reference.py python
scripts/fetch_gse270342_wheat_arabidopsis_orthologs.py python scripts/prepare_gse270342_wheat_root_case.py
python -m snowcell.cli train --config configs/gse270342_wheat_root_lora_adapter_4070.yaml --device cuda python
scripts/evaluate_checkpoint_detailed.py --config configs/gse270342_wheat_root_lora_adapter_4070.yaml --
checkpoint outputs/gse270342_wheat_root_lora_adapter_4070/best.pt --split test --output-dir
outputs/gse270342_wheat_root_lora_adapter_4070/detailed_test --device cuda --batch-size 16 python
scripts/promote_gse270342_wheat_adapter.py python scripts/audit_gse270342_wheat_lora_adapter.py python
scripts/run_scplantllm_gse270342_matched_baseline.py python
scripts/run_scplantllm_gse270342_partial_finetune.py python
scripts/audit_scplantllm_gse270342_partial_finetune.py python scripts/run_scplantllm_gse270342_full_finetune.py
--publish python scripts/audit_scplantllm_gse270342_full_finetune.py python
scripts/prepare_gse297576_bicolor_root_external.py --overwrite python
scripts/build_gse297576_sorghum_arabidopsis_ortholog_map.py python -m snowcell.cli annotate-bundle --checkpoint
models/SnowLotus_CellFM_SRP169576_annotation_1024_best.pt --data
outputs/external_validation/gse297576_bicolor_root/GSE297576_bicolor_root_author_atlas.h5ad --output-dir
outputs/external_validation/gse297576_bicolor_root/plantcellfm_frozen_bundle --ortholog-map
data/orthologs/gse297576_sorghum_to_arabidopsis_author_orthogroups.tsv --ortholog-aggregation first --batch-
size 64 --device cuda python scripts/audit_gse297576_bicolor_root_external.py python -m snowcell.cli train --
config configs/gse297576_sorghum_root_lora_adapter_4070.yaml --device cuda python
scripts/evaluate_checkpoint_detailed.py --config configs/gse297576_sorghum_root_lora_adapter_4070.yaml --
checkpoint outputs/gse297576_sorghum_root_lora_adapter_4070_oughw_holdout/best.pt --split test --output-dir
outputs/gse297576_sorghum_root_lora_adapter_4070_oughw_holdout/detailed_test --device cuda --batch-size 16
python scripts/promote_gse297576_sorghum_adapter.py python scripts/audit_gse297576_sorghum_root_adapter.py
python scripts/render_v7_sorghum_external_adaptation_figure.py python
scripts/audit_v7_sorghum_external_figure.py python scripts/build_v7_supplementary_tables.py python
scripts/render_v6_editorial_core_figures.py python scripts/render_v6_extended_evidence_figures.py python
scripts/audit_v6_submission_figure_suite.py
```

Figure Legends

Figure 1 | Plant-cell corpus contract and shared representation. A shared UMAP of 3,964 strict-evaluation cells is coloured by held-out species and ontology state. Corpus profile panels report the profiled five-species training record, strict-panel organ coverage and the gene-to-output annotation contract.

Figure 2 | Nested strict transfer retains open-set cells. The reference and strict output views use the same held-out cells. The denominator panel exposes all test cells, source-label-covered cells and open-set or unavailable labels. Species-level all-cell accuracy, conditional accuracy and coverage remain visible. The final panel compares centroid, expression, neural and context-aware transfer on the same 3,964 cells, exact-label denominator and 55.90% coverage. The last comparison is a global context-sensitivity analysis and does not replace the nested v17 primary result.

Figure 3 | Target-species adaptation dose response. Disjoint support and query cells are used at four labelled support budgets. Points are individual draws, bars are standard deviations, macro-F1 is reported separately and species-level values remain visible.

Figure 4 | Label-free external root execution and fixed-marker coherence. GSE152766/GSM4626007 is shown only with model outputs. The marker panel tests six literature-fixed root anchors against predicted groups. This figure is not an external accuracy or model-ranking result.

Figure 5 | Source-pinned external screen and sealed-library Sorghum adaptation. The GSE297576 author-labelled Sorghum bicolor root atlas is absent from the frozen five-species corpus. Frozen inference is completed before author labels are joined, and its complete comparable-cell denominator remains visible. A rank-8 LoRA adapter is trained on two libraries, selected on a third and evaluated once on the sealed OUGHW library. The main recovery panel compares frozen and adapted broad identities on the same 3,549 cells with fixed bootstrap intervals; paired UMAPs, all 27 raw author states and the feature-transfer audit provide compact traceability. This is target-species adaptation, not a zero-shot or third-party ranking.

Extended Data Figure 6 | Author-labelled secondary-root LoRA-mode adaptation. The frozen root checkpoint is adapted on the GSE270140/GSM8335426 train partition and selected on validation cells. The matched semantic recovery, 14-class held-out confusion matrix, per-class F1, validation history and locked-test metrics are shown. The protocol is supervised one-sample adaptation, not zero-shot or independent validation.

Extended Data Figure 7 | Source-only Arabidopsis-to-wheat transfer audit. A frozen root checkpoint and a GSE270140 source adapter are applied to a fixed three-state wheat evaluation after the source-only k=9 protocol is declared. The frozen checkpoint reaches 0.4231 macro-F1, whereas the source adapter reaches 0.4036; the negative adaptation result is retained. This is a source-only transfer sensitivity analysis, not the primary leave-species protocol or independent external validation.

Extended Data Figure 8 | Matched official scPlantLLM reference. The official scPlantLLM checkpoint uses the same GSE270342 prepared object, author first-target mapping and 1,433 locked test barcodes as the Plant-CellFM wheat adapter. The frozen centroid, final-block-plus-new-head partial adaptation and full-backbone-plus-new-head adaptation are shown with per-class changes and exact replay audits. This establishes a matched executable same-study reference, not independent validation, strict transfer, a compute-matched comparison or a universal ranking.

References

Chen, H. et al. scPlantDB: a comprehensive database for exploring cell types and markers of plant cell atlases. Nucleic Acids Research 52, D1629-D1638 (2024).

Zhai, J. et al. scPlant: a versatile framework for single-cell transcriptomic data analysis in plants. Plant Communications (2023).

Cui, H. et al. scGPT: toward building a foundation model for single-cell multi-omics. Nature Methods 21, 1470-1480 (2024).

Hao, M. et al. Large-scale foundation model on single-cell transcriptomics. Nature Methods 21, 1481-1491 (2024).

Rosen, Y. et al. Toward universal cell embeddings: integrating single-cell RNA-seq datasets across species with SATURN. Nature Methods 21, 1492-1500 (2024).

Theodoris, C. V. et al. Universal cell embedding provides a foundation model for cell biology. Nature (2026).

Lotfollahi, M. et al. Nicheformer: a foundation model for single-cell and spatial omics. Nature Methods 22, 2525-2538 (2025).

Jean-Baptiste, K. et al. Dynamics of gene expression in single root cells of Arabidopsis thaliana. Plant Cell 31, 993-1011 (2019).

Shahan, R. et al. A single-cell Arabidopsis root atlas reveals developmental trajectories in wild-type and cell identity mutants. Developmental Cell 57, 543-560.e9 (2022).

NCBI Gene Expression Omnibus. Arabidopsis thaliana GSE152766, sample GSM4626007.

Lyu, M. et al. The dynamic and diverse nature of parenchyma cells in the Arabidopsis root during secondary growth. Nature Plants (2025). https://doi.org/10.1038/s41477-025-01938-6

Ke, Y. et al. A single-cell and spatial wheat root atlas with cross-species annotations delineates conserved tissue-specific marker genes and regulators. Cell Reports 44, 115240 (2025). https://doi.org/10.1016/j.celrep.2025.115240

NCBI Gene Expression Omnibus. Triticum aestivum GSE270342: a soil-grown wheat root atlas with validated cross-species cluster annotations. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE270342

Submission Figure Suite

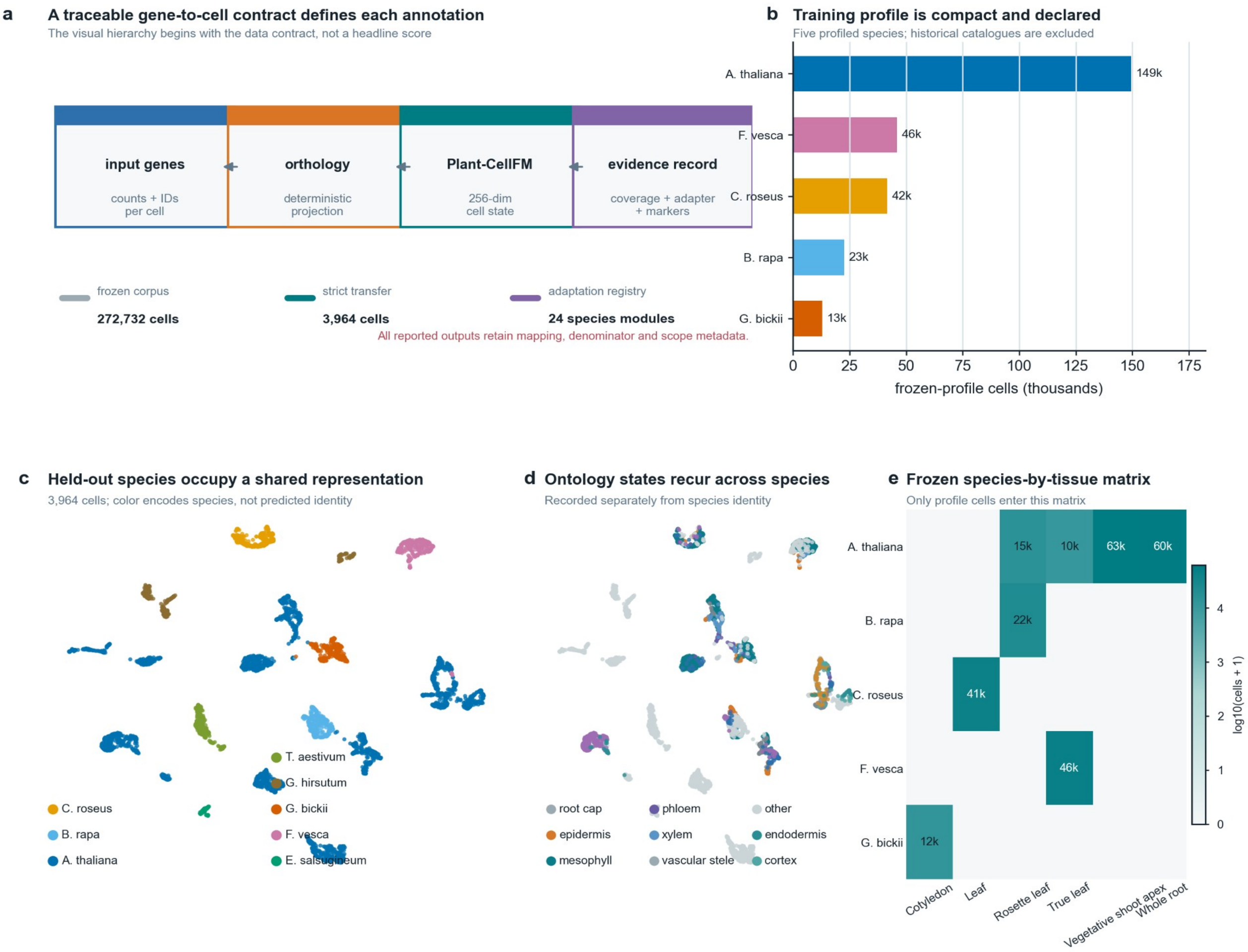


Figure 1 | Input contract, shared representation and auditable species-adaptation record.

Figure 2 | Nested leave-species transfer with the open-set denominator shown explicitly.

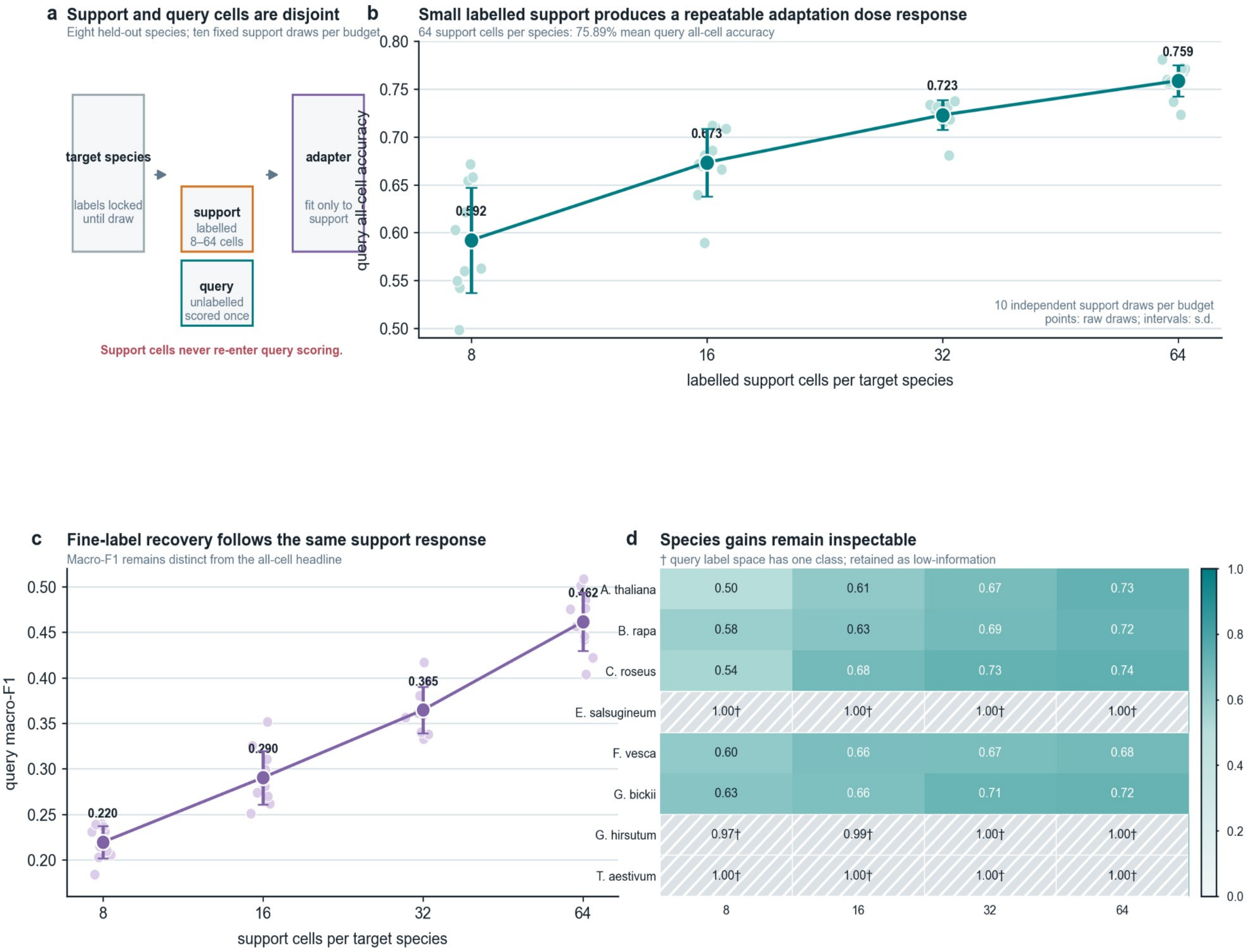


Figure 3 | Repeated target-labelled support response with strict support/query separation.

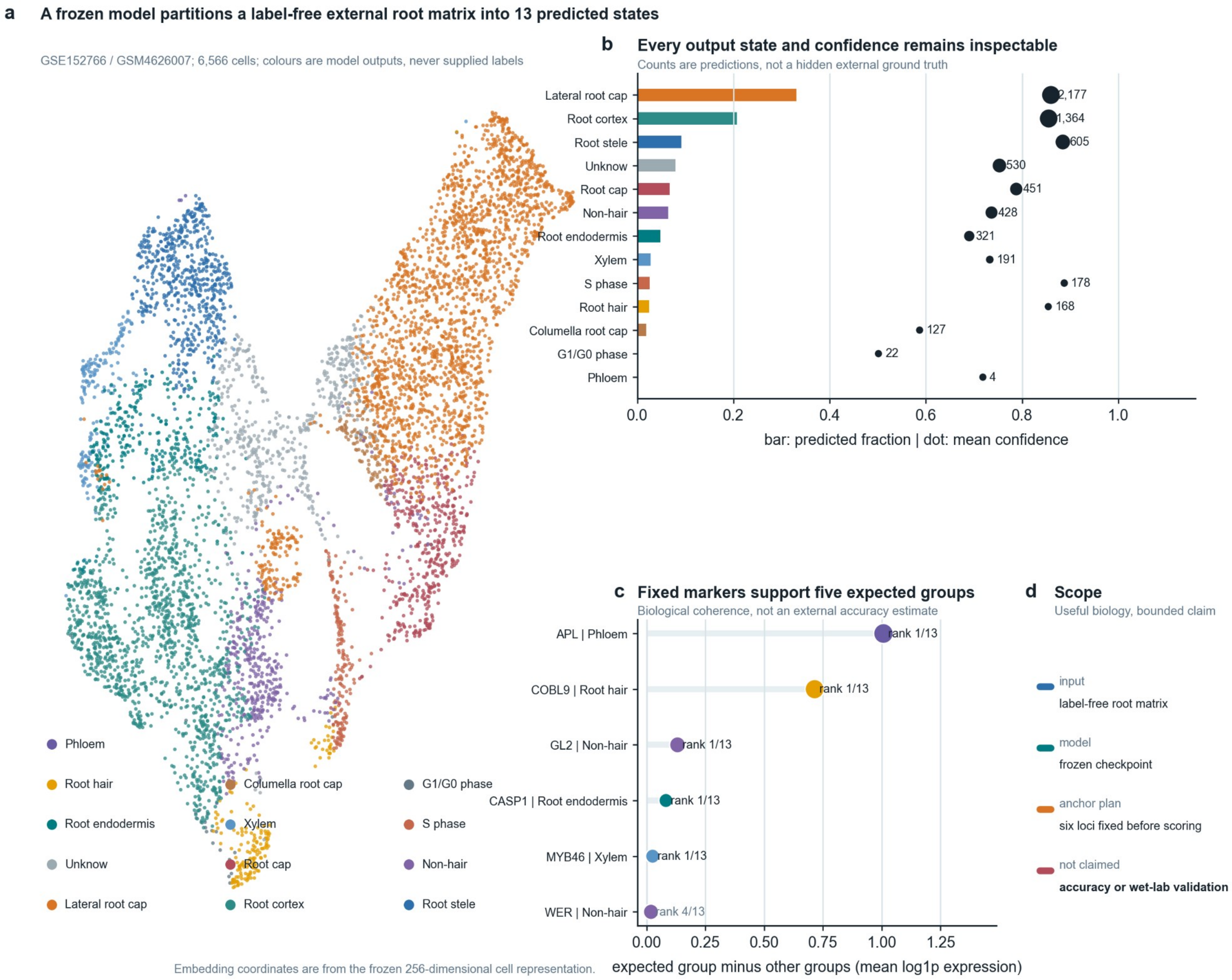


Figure 4 | Label-free external root execution and prespecified marker-coherence evidence.

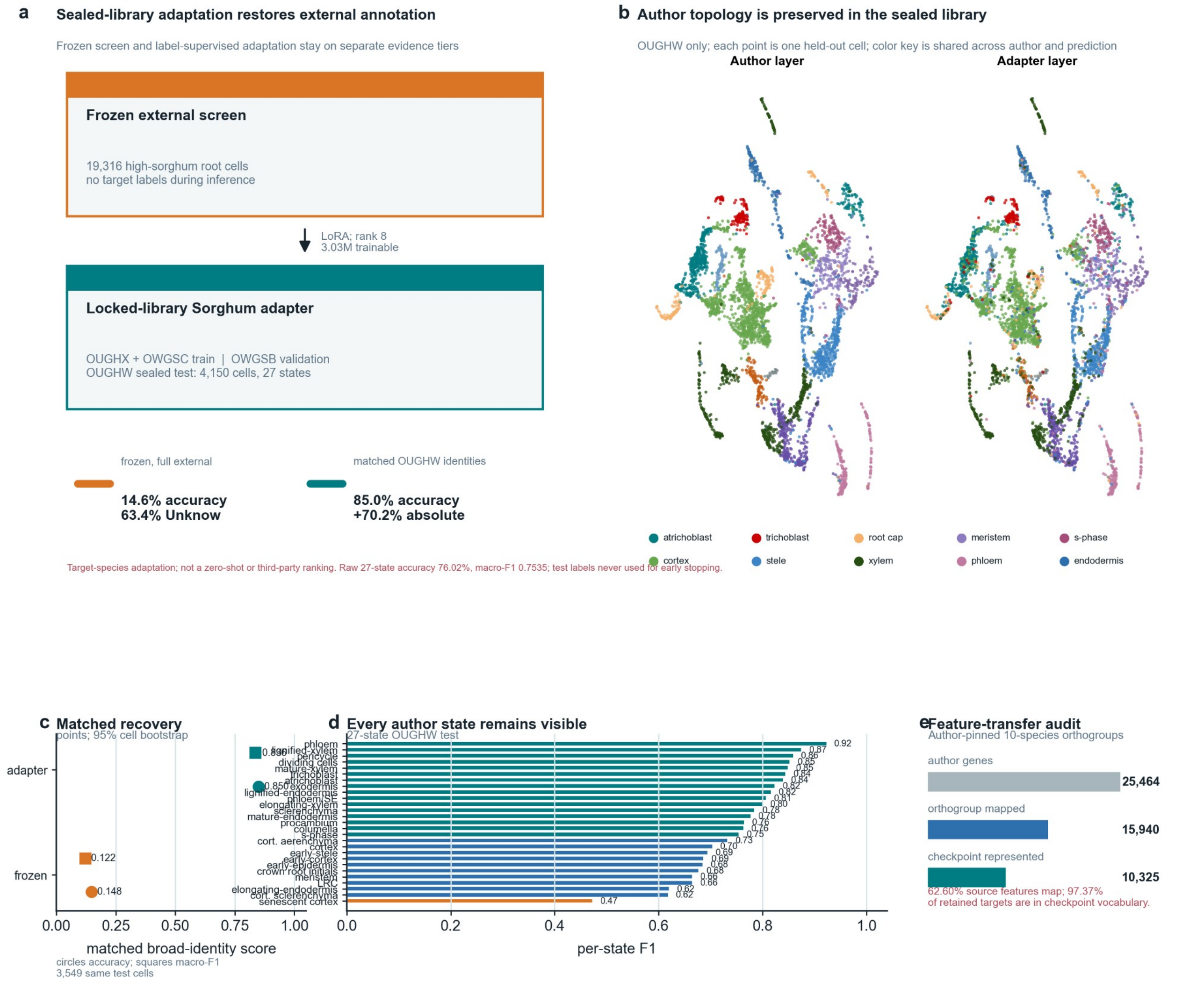
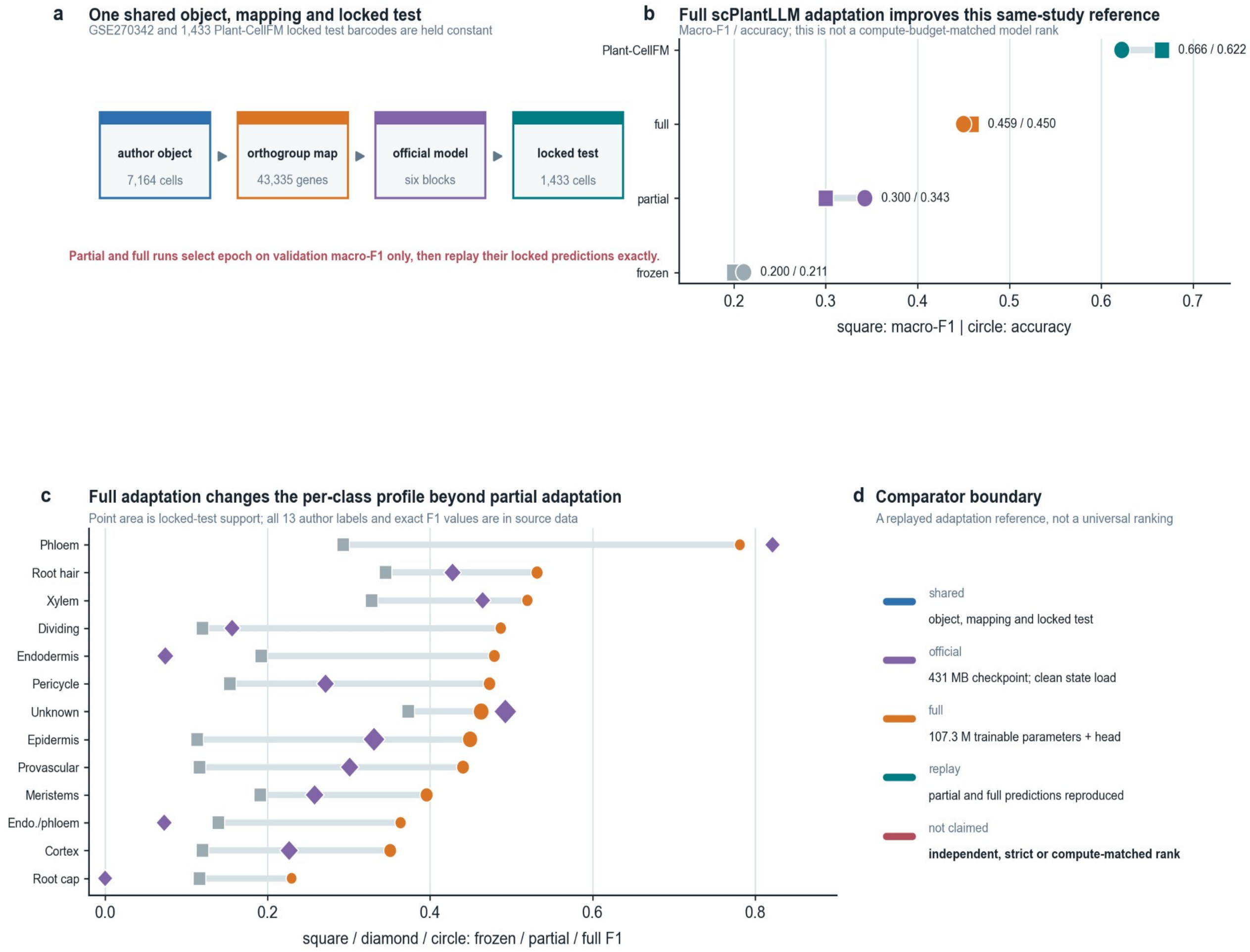


Figure 5 | Source-pinned external Sorghum screen and sealed-library recovery. The frozen result is species-absent and zero-shot; the high recovery is target-species supervised adaptation on the same sealed test library, not zero-shot or a third-party ranking.

Extended Data Figure 7 | Source-only Arabidopsis-to-wheat transfer audit, retaining the negative source-adapter result.



Extended Data Figure 8 | Official scPlantLLM reference on the same locked test, including frozen, partial-adaptation and full-backbone-plus-new-head paths. This is a same-study adaptation reference, not independent validation or a compute-matched rank.